

## A12-CPP HP, Netherlands

WMO: 062050

Lat: 55.399N

Lon: 3.810E

Elev: 108

StdP: 14.64

Time Zone: 1.00 (EUC)

Period: 09-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 31.8       | 33.4 | 20.3                          | 15.3 | 33.5 | 22.4 | 16.9 | 34.6 | 46.3                  | 43.1 | 42.7 | 43.2 | 23.1                  | 70   | 0.883 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 3.0                    | 66.9            | 63.6 | 65.5 | 62.4 | 64.4 | 61.5 | 64.4                | 66.1 | 63.0 | 64.8 | 61.9 | 63.9 | 15.5                 | 210  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 63.6                            | 88.2 | 65.6 | 62.2 | 84.0 | 64.5 | 60.9 | 80.2 | 63.5 | 29.6          | 66.4 | 28.6 | 64.9 | 27.8 | 64.0 | 68.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 41.1          | 37.3 | 33.9 | DB                         | 31.7 | 69.2               | 2.8 | 4.5                                                | 29.7 | 72.5       | 28.1 | 75.1       | 26.6 | 77.7       | 24.5 | 81.0 | (4) |
| (5)               |      |      | WB                         | 28.3 | 64.9               | 3.2 | 2.4                                                | 26.0 | 66.7       | 24.1 | 68.1       | 22.3 | 69.4       | 20.0 | 71.2 | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |                                                                                  | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |      |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                                  |                                                                                  | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |      |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg                                                                            | 50.1   | 41.8  | 40.6  | 41.8  | 44.9  | 49.8  | 54.8  | 59.9  | 61.2  | 58.5  | 54.1  | 48.8  | 44.8 |      |
|                                                                                  | DBStd                                                                            | 7.90   | 3.58  | 3.76  | 3.60  | 3.21  | 3.60  | 2.85  | 3.20  | 2.62  | 2.76  | 3.15  | 4.00  | 3.84 |      |
|                                                                                  | HDD50                                                                            | 1211   | 255   | 263   | 253   | 157   | 48    | 1     | 0     | 0     | 0     | 7     | 64    | 163  |      |
|                                                                                  | HDD65                                                                            | 5429   | 720   | 683   | 718   | 604   | 472   | 305   | 162   | 119   | 195   | 338   | 485   | 628  |      |
|                                                                                  | CDD50                                                                            | 1263   | 0     | 0     | 0     | 2     | 41    | 146   | 308   | 348   | 255   | 133   | 29    | 1    |      |
|                                                                                  | CDD65                                                                            | 7      | 0     | 0     | 0     | 0     | 0     | 0     | 4     | 2     | 1     | 0     | 0     | 0    |      |
|                                                                                  | CDH74                                                                            | 1      | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0     | 0    |      |
|                                                                                  | CDH80                                                                            | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |      |
| Wind                                                                             | WSAvg                                                                            | 18.3   | 21.6  | 20.6  | 18.9  | 16.3  | 16.4  | 14.9  | 14.7  | 15.5  | 17.2  | 20.5  | 20.6  | 22.0 |      |
| Precipitation                                                                    | PrecAvg                                                                          | 44.6   | 5.5   | 4.4   | 3.7   | 2.2   | 1.8   | 2.4   | 2.8   | 3.5   | 3.9   | 4.3   | 5.1   | 5.1  |      |
|                                                                                  | PrecMax                                                                          | 55.3   | 9.6   | 7.3   | 6.5   | 3.9   | 2.7   | 4.3   | 4.0   | 5.7   | 6.5   | 6.2   | 8.1   | 9.9  |      |
|                                                                                  | PrecMin                                                                          | 36.1   | 1.9   | 1.9   | 1.3   | 0.9   | 0.9   | 0.9   | 1.3   | 1.5   | 1.8   | 1.7   | 3.2   | 2.5  |      |
|                                                                                  | PrecStd                                                                          | 4.4    | 1.9   | 1.5   | 1.2   | 0.7   | 0.5   | 0.7   | 0.8   | 0.9   | 1.1   | 1.2   | 1.2   | 1.7  |      |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%                                                                             | DB     | 49.3  | 48.2  | 49.3  | 53.9  | 60.1  | 63.6  | 71.0  | 68.3  | 67.1  | 62.1  | 58.0  | 52.1 |      |
|                                                                                  |                                                                                  | MCWB   | 46.9  | 45.4  | 46.3  | 48.4  | 56.1  | 59.8  | 66.8  | 64.8  | 64.3  | 59.0  | 55.6  | 49.4 |      |
|                                                                                  | 2%                                                                               | DB     | 48.2  | 46.8  | 47.7  | 51.4  | 57.8  | 61.2  | 67.5  | 66.5  | 65.3  | 60.6  | 56.2  | 51.0 |      |
|                                                                                  |                                                                                  | MCWB   | 45.9  | 44.5  | 45.5  | 47.8  | 54.6  | 58.5  | 64.3  | 63.0  | 62.5  | 58.1  | 53.9  | 48.7 |      |
|                                                                                  | 5%                                                                               | DB     | 47.1  | 45.8  | 46.6  | 50.1  | 55.8  | 59.9  | 65.5  | 65.6  | 63.5  | 59.4  | 55.0  | 50.0 |      |
|                                                                                  |                                                                                  | MCWB   | 44.9  | 43.8  | 44.6  | 47.3  | 53.3  | 57.5  | 62.5  | 62.2  | 60.8  | 56.7  | 52.6  | 48.1 |      |
|                                                                                  | 10%                                                                              | DB     | 46.3  | 45.1  | 45.7  | 48.9  | 54.3  | 58.4  | 64.1  | 64.7  | 62.3  | 58.1  | 53.9  | 49.1 |      |
|                                                                                  |                                                                                  | MCWB   | 44.0  | 43.1  | 43.7  | 46.7  | 51.8  | 56.0  | 61.3  | 61.6  | 59.5  | 55.4  | 51.4  | 47.0 |      |
|                                                                                  | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%   | WB    | 47.7  | 46.5  | 46.8  | 49.6  | 57.3  | 60.4  | 67.3  | 65.9  | 64.6  | 59.9  | 55.8 | 50.4 |
|                                                                                  |                                                                                  |        | MCDB  | 48.7  | 47.4  | 48.4  | 52.2  | 59.0  | 62.3  | 69.7  | 67.3  | 65.7  | 61.3  | 57.2 | 51.0 |
| 2%                                                                               |                                                                                  | WB     | 46.5  | 45.2  | 45.9  | 48.5  | 55.5  | 59.1  | 65.1  | 64.0  | 63.0  | 58.6  | 54.5  | 49.4 |      |
|                                                                                  |                                                                                  | MCDB   | 47.7  | 46.2  | 47.3  | 50.5  | 56.8  | 60.6  | 67.3  | 65.6  | 64.3  | 60.5  | 55.9  | 50.4 |      |
| 5%                                                                               |                                                                                  | WB     | 45.5  | 44.3  | 44.9  | 47.7  | 53.6  | 57.9  | 63.1  | 63.0  | 61.3  | 57.3  | 53.1  | 48.5 |      |
|                                                                                  |                                                                                  | MCDB   | 46.7  | 45.4  | 46.2  | 49.5  | 55.4  | 59.6  | 65.0  | 64.8  | 62.9  | 58.9  | 54.8  | 49.8 |      |
| 10%                                                                              |                                                                                  | WB     | 44.5  | 43.5  | 44.1  | 46.8  | 52.1  | 56.4  | 61.6  | 62.1  | 60.2  | 56.1  | 51.6  | 47.4 |      |
|                                                                                  |                                                                                  | MCDB   | 45.9  | 44.8  | 45.4  | 48.6  | 53.9  | 58.2  | 63.7  | 64.1  | 61.9  | 57.9  | 53.5  | 48.9 |      |
| Mean Daily<br>Temperature<br>Range                                               |                                                                                  | 5% DB  | MDBR  | 3.7   | 3.4   | 3.3   | 3.3   | 3.3   | 2.8   | 3.1   | 3.0   | 3.1   | 3.5   | 3.8  | 3.8  |
|                                                                                  |                                                                                  |        | MCDBR | 4.1   | 3.8   | 3.8   | 4.6   | 5.3   | 4.2   | 4.2   | 3.7   | 4.0   | 4.1   | 4.3  | 3.8  |
|                                                                                  | MCWBR                                                                            |        | 5.2   | 3.6   | 3.6   | 3.0   | 3.7   | 3.6   | 3.5   | 4.3   | 4.8   | 4.7   | 5.1   | 4.7  |      |
|                                                                                  | 5% WB                                                                            | MCDBR  | 3.9   | 3.6   | 3.6   | 4.4   | 5.1   | 4.0   | 4.0   | 3.5   | 3.8   | 4.3   | 4.3   | 3.9  |      |
|                                                                                  |                                                                                  | MCWBR  | 5.1   | 3.5   | 3.7   | 3.1   | 3.9   | 3.7   | 4.0   | 4.5   | 4.5   | 5.1   | 5.4   | 5.0  |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub                                                                             | 0.335  | 0.348 | 0.369 | 0.401 | 0.383 | 0.389 | 0.393 | 0.391 | 0.380 | 0.356 | 0.336 | 0.316 |      |      |
|                                                                                  | taud                                                                             | 2.240  | 2.316 | 2.313 | 2.260 | 2.402 | 2.402 | 2.404 | 2.414 | 2.426 | 2.399 | 2.296 | 2.234 |      |      |
|                                                                                  | Ebn at Noon                                                                      | 179    | 227   | 252   | 259   | 273   | 272   | 268   | 261   | 247   | 222   | 178   | 157   |      |      |
|                                                                                  | Edh at Noon                                                                      | 19     | 25    | 31    | 37    | 34    | 35    | 34    | 32    | 28    | 23    | 17    | 15    |      |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg                                                                           | 156    | 367   | 793   | 1344  | 1773  | 1783  | 1707  | 1419  | 941   | 492   | 203   | 115   |      |      |
|                                                                                  | RadStd                                                                           | 19     | 53    | 72    | 115   | 123   | 115   | 172   | 98    | 111   | 35    | 24    | 14    |      |      |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (8 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page